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Thermoacoustic refrigeration with compressible liquid as working fluid

Réfrigération thermoacoustique avec liquide compressible comme fonctionnement fluide

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Highlights

- Both an idealized and a full-size refrigerator using liquid are modeled.
- The liquid thermoacoustic refrigeration may be as efficient as the gas system .
- Both the linear motor and the heat exchanger can be much more compact.

Abstract

The classical thermoacoustic refrigeration adopts gases with ideal-gas properties as working fluids. Herein, the thermoacoustic refrigeration using a compressible liquid as the working fluid is investigated for the first time. Two models are utilized parallelly: a highly-simplified short refrigerator model and a far-more realistic full-scale thermoacoustic refrigerator model. The results from the short refrigerator model show that, ideally, the coefficient of performance can reach above 70% of the Carnot limit. In the full-scale system discussion, the coefficient of performance of the system can reach as high as 4.1 in a typical working condition for air conditioning (pumping heat from 285 K to 310 K), with respect to 36% of the Carnot limit. When pumping

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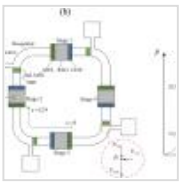
heat from 332 K to 357 K, the coefficient of performance can reach 44% of the Carnot limit. These results indicate that the efficiency of the thermoacoustic refrigeration with a compressible liquid as the working fluid is potentially high, but with a much smaller displacement amplitude of the fluid. As a result, both the acoustic source (the linear motor, for instance) and the heat exchanger can be much more compact, so that the mass and the volume of the system can be greatly reduced, as well as the cost.

Keywords

Thermoacoustic refrigeration; Liquid thermoacoustic refrigeration; Thermoacoustic heat pump; Critical point

Mots clés

Réfrigération thermoacoustique; Réfrigération thermoacoustique liquide; Pompe à chaleur thermoacoustique; Point critique



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Table 1. Dimensions of com...

Nomenclature

English letters

A
: Area, m^2

a
: sound velocity, m/s

c_p
: isobaric heat capacity, J/kg·K

COP
: coefficient of performance of the short refrigerator

COP^s
: coefficient of performance of the full-scale system

COP_R
: relative Carnot COP of the short refrigerator

COP_R^s
: relative Carnot COP of the full-scale system

$\dot{E}_2$

acoustic power, W

f

frequency, Hz

f_κ

spatially averaged thermal function

f_v

spatially averaged viscous function

g

acoustic source term, m^{-1}

$\dot{H}_2$

rate of total energy flux, W

h

heat transfer coefficient, $W/m^2 \cdot K$

k

thermal conductivity, W/m·K

L

total length of the loop, m

l_{hx}

length of heat exchanger, m

p

pressure, Pa

Pr

Prandtl number

$\dot{q}$

gradient of $\dot{H}_2$, W/m

Q

rate of heat exchange through heat exchangers, W

r_h

hydraulic radius, m

r_κ

thermal-relaxation resistance per unit length, $kg/m^3 \cdot s$

r_v

viscous resistance per unit length, $kg/m^5 \cdot s$

T

temperature, K

U

volumetric velocity, m^3/s

x

axial direction, m

x_{eff}

effective length of heat transfer, m

z

acoustic impedance, defined as p_1 / u_1, $kg/m^2 \cdot s$

Greek letters

β

thermal expansion coefficient, K^{-1}

δ_κ

thermal penetration depth defined as $\sqrt{2\kappa / \omega}$, m

ϵ_s

correction for thermal properties of the solid wall

γ

specific heat ratio

κ

thermal diffusivity, m^2/s

ω

angular frequency, s^{-1}

$\omega\tau$

Womersley number, defined as $\frac{r_h^2\omega}{2\kappa}$

ρ

density, kg/m^3

φ

phase lead of p_1 relative to U_1, deg.

ξ

displacement of fluid, m

Subscripts

1

first order

a
: through ambient heat exchanger

ac
: acoustic source

as
: for acoustic source

c
: through cold heat exchanger

cr
: critical point

fs
: between fluid and solid in heat exchanger

gas
: for gas

he
: for helium

in
: inlet

lq
: for liquid prolylene

m
: mean value

out
: outlet

solid
: for solid

Other symbols

$\Im$
: imaginary part of a complex number

$\Re$
: real part of a complex number

$\sim$
: conjugate of a complex number

ph

phase angle of a complex number

AHX
: ambient heat exchanger

CHX
: cold heat exchanger

i
: unit of imaginary number

TBT
: thermal buffer tube

1. Introduction

A thermoacoustic refrigerator is a device that consumes acoustic power to pump heat from a low-temperature heat source to a high-temperature environment through thermoacoustic effect. It generally has four main components: a stack/regenerator for thermoacoustic conversion, heat exchangers for heat transfer and creating a temperature gradient along the stack/regenerator, ducts to create proper acoustic field and a sound source to drive the system. Its extremely simple structure endows a thermoacoustic refrigerator with advantages of high reliability and low cost. Thus the thermoacoustic refrigerator has been used in space applications (Garrett et al., 1993), household refrigeration (Chi et al., 2021, Chi et al., 2022), electronic device cooling (Chen and Xu, 2022) and natural gas liquefaction (Wang et al., 2022, Wang et al., 2023). Generally, gases (helium, nitrogen and argon, etc.) with ideal-gas behaviors are adopted in thermoacoustic refrigerators. However, conventional thermoacoustic refrigerators still suffer from problems like low energy density (Chen et al., 2021).

Limited by daily experience, people often have a false idea that liquids cannot be compressed and thus is thermodynamically inert. Consequently, a liquid has rarely been considered as the working fluid for heat engines. However, a liquid may become a very good working fluid under specific conditions. Let us first consider $T\beta$, which is an important parameter to estimate whether a fluid is proper for thermoacoustic conversion because it affects the conversion directly (the acoustic source term is proportional to it). For a fluid in the range of $0.7T_{cr} \lesssim T \lesssim T_{cr}$ and $p > p_{cr}$, the value of $T\beta$ can be close to or even beyond unity, which is the value for ideal gases, otherwise $T\beta$ may be too low to

be used for thermoacoustic systems. Therefore, to ensure the thermodynamic activity, a liquid is only suitable for low temperature difference applications including waste heat recovery and air conditioning (Swift, 1989b). Secondly, heat capacity ρc_p of liquids is much larger than that of gases, leading to a much higher heat transfer coefficient (Allen et al., 1980). Lastly, the characteristic acoustic impedance $\rho_m a$ of liquids is orders of magnitude higher than that of gases, which helps the parcels to oscillate with a large pressure amplitude but a small velocity amplitude. This significantly reduces the risk of explosion, and allows a high-power system with a very small stroke (Allen et al., 1981b).

The idea of using a liquid as the working fluid in heat engines was firstly proposed by J.F.J Malone about one hundred years ago (Malone, 1929, Malone, 1924). In the engines he proposed, the piston and the displacer move back and forth repeatedly, while the fluid inside the heat exchangers and the regenerator moves in a unidirectional pulsating way with the help of check valves. In one of his engines, liquid water with pressure as high as 700 bar was used as the working fluid. A thermal efficiency of 20% was achieved, with a brake power up to 368 kW (500 horse power) (Malone, 1931). This performance was superior to steam engines for powering ships and locomotives during that era. In fact, Malone was confident with the commercial application of his engine, but he could not realize this due to the Great Depression and the rapid development of other engines (internal combustion engine, for instance).

The above-mentioned engine was called Malone engine, which had been unfortunately forgotten for a long time, until a group from the University of California led by Wheatley (Allen et al., 1981a, Allen et al., 1980, Allen et al., 1981b) rediscovered this idea in 1980s. They realized that the nature of Malone engine is a Stirling-cycle engine but with a liquid as the working fluid, although with characteristics of Brayton cycle. Later, Swift (1989b) developed a Stirling system with liquid propylene under a mean pressure of 102 bar as the working fluid. It can be considered as four alpha-type Stirling engines connected in series by a common crankshaft, with a phase difference of 90° between adjacent sub-engines. To reduce the viscous loss, the regenerator was multi-layer stacked metal sheets composed of spiraled 0.15-mm-thick stainless-steel films, with a gap of 40 μm between adjacent layers. The gap was controlled by the bumps on the

sheet (Swift et al., 1983). The system could operate as either an engine or a refrigerator. In the refrigerator mode, it could obtain a coefficient of performance of 1.8 (corresponding to 20% of the Carnot limit) when pumping heat with a temperature difference of around 30 K to the ambient temperature environment (Swift, 1989b, Swift, 1989a).

Given the essential similarity between thermoacoustic and Stirling systems (Ceperley, 1979), a liquid thermoacoustic engine has also been proposed by Migliori and Swift (1988). Liquid sodium was used as the working fluid. The engine produced 18 W acoustic power with a thermal efficiency of 1.8%. Moreover, liquid CO_2 was also suggested as the working fluid (Swift and Brown, 1994), but no further study was carried out. Recently, a research group led by Scalo in Purdue University reported a series of theoretical and experimental studies on thermoacoustic engines using fluids near critical points, observing significantly reduced temperature difference to drive the engine and a very high pressure amplitude (Migliorino and Scalo, 2020, Scalo et al., 2017, Martinez et al., 2021a, Martinez et al., 2021b, Alexander et al., 2018). However, their works are on engines with trans-critical fluids.

Motivated by the above mentioned studies, the compressible liquid might be a promising working fluid for thermoacoustic refrigeration with its unique advantages. However, up until now,the liquid thermoacoustic refrigeration has yet to be studied in details, and its mechanism and potential in performance remain unclear. Herein, a systematic theoretical study on the liquid thermoacoustic refrigeration, especially in the traveling-wave acoustic field, is presented. Both a highly-simplified short refrigerator model and a far-more realistic full-scale thermoacoustic refrigerator model are used. Our results demonstrate that the liquid thermoacoustic refrigeration can be efficient, towards the conventional gas thermoacoustic refrigeration in low-temperature lift applications, and with a more compact configuration.

2. Model formulation

2.1. Governing equations

The theory adopted in this work is based on Rott's thermoacoustic theory, but with some modifications to fit real fluids (Swift et al., 1985). Specifically, the governing equations

can be expressed as :

$$\frac{dp_1}{dx} = -\frac{i\omega\rho_m}{1-f_v A_{\text{gas}}} U_1 \tag{1}$$

$$\frac{dU_1}{dx} = -\frac{i\omega A_{\text{gas}}}{\rho_m a^2}\left[1+\gamma-1 f_\kappa\right] p_1 + \frac{\beta f_\kappa - f_v}{1-f_v(1-Pr)}\frac{dT_m}{dx} U_1 \tag{2}$$

$$\frac{dT_m}{dx} = \frac{\dot{H}_2 - \frac{1}{2}\Re\left[\tilde{p}_1 \tilde{U}_1^*\right]\left[1-\frac{T_m\beta f_\kappa - \tilde{f}_v}{1+\epsilon_s(1+Pr)(1-\tilde{f}_v)}\right]}{\frac{\rho_m c_p |U_1|^2}{2\omega A_{\text{gas}}(1-Pr)|1-f_v|^2}\Im\tilde{f}_v + \frac{f_\kappa - \tilde{f}_v}{1+\epsilon_s(1+Pr)}\frac{1+\epsilon_s f_v/f_\kappa}{} - A_{\text{gas}}k - A_{\text{solid}}k_{solid}} \tag{3}$$

$$\frac{d\dot{H}_2}{dx} = \dot{q} \tag{4}$$

Eqs. (1) to (3) describe the balances of the momentum, the mass continuity and the total energy flux respectively. Eq. (4) shows the energy exchange between the thermoacoustic system and the environment. $\frac{d\dot{H}_2}{dx}$ is zero in every component except in heat exchangers.

$\dot{H}_2$ is the total energy flux, which is defined as

$$\dot{H}_2(x) = \frac{1}{2}\rho_m c_p \int \Re T_1 \tilde{u}_1 dA + \frac{1}{2}(1-\beta T_m)\int \Re p_1 \tilde{u}_1 dA - Ak + A_{\text{solid}}k_{\text{solid}}\frac{dT_m}{dx} \tag{5}$$

Additionally, f_κ and f_v are spatial-averaged functions for thermal and viscous effects. For parallel-plate regenerators, their expressions are (Swift, 2002)

$$f_\kappa = \frac{\tanh(1+i)\sqrt{\omega\tau}}{(1+i)\sqrt{\omega\tau}}, \tag{6a}$$

$$f_v = \frac{\tanh(1+i)\sqrt{\omega\tau}/\sqrt{Pr}}{(1+i)\sqrt{\omega\tau}/\sqrt{Pr}} \tag{6b}$$

where $\omega\tau \equiv (r_h/\delta_\kappa)^2$, with $\delta_\kappa \equiv \sqrt{2\kappa/\omega}$ being the thermal penetration depth. Apparently, f_κ and f_v are frequency-dependent. Eqs. (1) to (4) can be used to depict the motion of the liquid in all components of a thermoacoustic system. These equations retreat to those for the ideal gas when $\beta = \frac{1}{T_m}$.

2.2. Model system

2.2.1. Short refrigerator

We first propose a short refrigerator model. As shown in Fig. 1(a), the short refrigerator is actually a short parallel-plate type regenerator, which is the only place where thermoacoustic conversion happens in a traveling-wave thermoacoustic system, with the length of Δx. It is so short that the values of p_1, U_1, T_m

and the properties of the fluid inside it can be treated constant. Then the influence of the liquid on the performance can be decoupled from the acoustic field and the geometry of the system. Therefore, this idealized model can be used to investigate the full potential of a specific thermoacoustic cycle, regardless of the geometry of the other components in a full-scale system (Swift and Fusco, 1988, Yang et al., 2022a). Additionally, this short structure is called a refrigerator, because it pumps heat from its lower-temperature end to its higher-temperature end, with the consumption of acoustic power. This actually follows the idea of the "short engine" in Swift (1988).

The acoustic power consumed by the short refrigerator can be calculated by

$$\dot{\Delta E}_2 = -\frac{\Delta x}{2}\Re p_1 \frac{d\tilde{U}_1}{dx} + \frac{dp_1}{dx}\tilde{U}_1 \tag{7}$$

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Fig. 1. (a) The short refrigerator model. It can be considered a short section of the regenerator with the length of Δx. (b) The four-stage traveling-wave thermoacoustic refrigerator with liquid as the working fluid. (c) Regime of compressible liquid and conventional thermoacoustic conversions in p-T diagram. Typically, the compressible liquid covers the pressure range of $p > p_{cr}$. TA-thermoacoutsics. AHX-ambient heat exchanger. REG-regenerator. CHX-cold heat exchange. TBT-thermal buffer tube. 2nd AHX-secondary ambient heat exchanger.

From the energy balance in the regenerator, the cooling power can be calculated through (Swift, 2002)

$$\dot{Q}_c = \dot{E}_{2,c} - \dot{H}_2 = -\frac{dT_m}{dx}\frac{\rho_m c_p |U_1|^2}{2\omega A_{\text{gas}}(1-Pr)|1-f_v|^2}\Im\tilde{f}_v + \frac{f_\kappa - \tilde{f}_v 1 + \epsilon_s f_v / f_\kappa}{1+\epsilon_s(1+Pr)} - A_{\text{gas}}k - A_{\text{solid}}k_s + \frac{1}{2}\Re p_1 \tilde{U}_1 \frac{T_m \beta f_\kappa - \tilde{f}_v}{1+\epsilon_s(1+Pr)1-} \tag{8}$$

Thus, the coefficient of performance can be expressed as

$$COP = \frac{\dot{Q}_c}{\dot{\Delta E}_2} \tag{9}$$

Hence, the relative Carnot COP is

$$COP_R = \frac{COP}{COP_{carnot}} = \frac{\Delta T_m COP}{T_m} \tag{10}$$

By substituting Eqs. (8), (9) into Eq. (10) and ignoring the conduction loss in the axial direction, we reach

$$COP_R = F p_m, T_m, \nabla T_m / \omega, |z|, \varphi, \omega\tau, Fluid \quad (11)$$

In this work we focus on the traveling-wave thermoacoustic conversion, which means ideally the conversion happens in a traveling-wave field, so φ is fixed at zero. Thus, COP_R depends on six factors, which are mean pressure p_m, temperature T_m, the group $\nabla T_m / \omega$, magnitude of acoustic impedance $|z|$, Womersley number $\omega\tau$ and fluid properties.

2.2.2. Full-scale traveling-wave refrigerator

To give a more practical estimation of the performance of the liquid thermoacoustic refrigeration, we also propose a four-stage traveling-wave thermoacoustic refrigerator. As can be seen in Fig. 1(b), the system has four identical stages, connected in series with a phase delay of 90°. In each stage, there is a regenerator sandwiched by an ambient heat exchanger (AHX) and a cold heat exchanger (CHX). A second AHX separates the tube connecting two stages into two sections, one is the thermal buffer tube (TBT) with a temperature gradient along it, the other one is the resonator which has no temperature gradient. This design is chosen because it is a mature design for classical thermoacoustic systems, able to provide a stable acoustic field dominated by traveling-wave components (Blok, 2012, Zhang et al., 2016). The system was designed with the thermoacoustic generator developed by Bi et al. (2017) as the reference. The details of each component can be seen in Table 1. As for the hydraulic radius of the regenerator, it varies around 10 μm, to satisfy a specific $\omega\tau$. Additionally, the acoustic source can be either a linear motor or a loudspeaker, or any other devices that can provide desired acoustic oscillations. It is treated as a black box. The frequency is set to be 60 Hz, with considerations of a proper frequency for the operation of linear motors (assume that they are used as acoustic sources) (Timmer et al., 2018) and the hydraulic radius to be not too small. At the joint of the acoustic source and the trunk resonator, U_1 is discontinuity and the Kirchhoff's law is obeyed, i.e., $U_{1,out} = U_{1,in} + U_{1,as}$ (see Fig. 1(b)), while p_1 stays unchanged.

The numerical methodology of this model is presented in Fig. 2. First, the configuration of the system is defined by specifying geometric parameters, and the operation parameters including the working fluid, frequency, mean pressure and ambient

temperature are also given. Then the governing equations (i.e., Eqs. (1) to (4)) can be integrated along the axial direction with a forth-order Runge–Kutta method, from $x = 0$ to $x = L / 4$. $\left|p_1\right|$ at $x = 0$ is fixed at a given value. We note that there are six unknown parameters, which are $\Re[U_1]$ and $\Im[U_1]$ at $x = 0$, $\Re[U_1]$ and $\Im[U_1]$ contributed by the acoustic source, Q_a and Q_c. In order to integrate the equations, the unknown parameters must be initialized with arbitrary values. To close the problem, six targets, or boundary conditions, need to be satisfied:

$$\left|p_1\right|_{x=0} = \left|p_1\right|_{x=L/4} \tag{12a}$$

$$\left|U_1\right|_{x=0} = \left|U_1\right|_{x=L/4} \tag{12b}$$

$$php_{1_{x=0}} = php_{1_{x=L/4}} + 90° \tag{12c}$$

$$phU_{1_{x=0}} = phU_{1_{x=L/4}} + 90° \tag{12d}$$

$$T_{m,x=0} = T_{m,x=L/4} \tag{12e}$$

$$T_c = \text{target value} \tag{12f}$$

Table 1. Dimensions of components of the thermoacoustic refrigerator.

Item	Length (mm)	Diameter (mm)	Hydraulic radius (mm)
AHX	2	170	0.2
Regenerator	35	170	varies to satisfy specific $\omega\tau$, typically ~10 μm
CHX	2	170	0.2
Resonator	1121	26	6.5
2nd AHX	2	26	0.5
TBT	640	26	6.5

Following the famous software DeltaEC (Clark et al., 2007), the unknown parameters are updated through a shooting algorithm until converged solutions are reached (Powell, 1970). As a result, the distributions of p_1, U_1 and T_m can be achieved, as well as other parameters to evaluate the performance of the refrigerator.

Thus, the coefficient of performance of the system can be expressed as

$$COP^s = \frac{Q_c}{E_{as}} \tag{13}$$

where E_{as} is the acoustic power delivered from the acoustic source.

Hence, the relative Carnot COP is

$$COP_R^s = \frac{COP}{COP_{carnot}} \tag{14}$$

In both models, although in a few calculations the system enters the supercritical region to better show the tendency of the performance, pressure and temperature of the system are primarily limited to $p > p_{cr}$ and $T < T_{cr}$, because the compressible liquid typically falls in this region (see Fig. 1(c)). Liquid propylene is used as the working fluid, because its critical point ($p_{cr} = 4.555$ MPa, $T_{cr} = 364.2$ K) is adequate for a refrigerator system working under compressible liquid region, and is feasible to be achieved in a lab environment. All the properties of fluid are obtained from NIST REFPROP (Lemmon et al., 2007).

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Fig. 2. Methodology of the full-scale model calculation.

3. Results and discussion

3.1. Properties of liquid

The uniqueness of the liquid thermoacoustic refrigeration comes from fluid properties. We exemplify this with propylene in Fig. 3. As can be seen, β goes to divergence at the critical point. There is also a peak for each curve when $p_m > p_{cr}$, but its value decreases as p_m is increased. Obviously, in a wide temperature range ($T_m >$ 300 K) when $p_m > p_{cr}$ which covers the requirement of air conditioning and some industrial heat pumps, β of propylene can be comparable to or even larger than that of the ideal gas. It implies that a liquid can be thermodynamically powerful for thermoacoustic conversion, lying the foundation of an efficient liquid thermoacoustic refrigerator. β gradually decreases when $T_m < 250$ K, because the fluid is approaching towards incompressible, which is not suitable as a working fluid. A smaller Pr is also important, which means a lower viscous loss relative to the thermal effect (Tijani et al., 2002). The trends of Pr are similar to those of β. However, a liquid always has $Pr > 1$,

while $Pr < 1$ for most gases. This is an obstacle when adopting a liquid. Additionally, Pr goes to very high near critical point, indicating severe viscous loss. The trends of ρ is also shown. There is a sudden decrease near critical point when T_m is lowered.

To illustrate a more straightforward understanding on how the unique properties affect thermoacoustic conversion, we reconsider the thermoacoustic conversion in the regenerator (or the short refrigerator). By rearranging Eqs. (1), (2), the variation of acoustic power along the axial direction can be expressed as (Swift, 2002)

$$\frac{dE_2}{dx} = -\frac{r_v}{2}\left|U_1\right|^2 - \frac{1}{2r_\kappa}\left|p_1\right|^2 + \frac{1}{2}\Re\tilde{gp}_1 U_1 \tag{15}$$

where the three terms represent the viscous loss, the thermal-relaxation loss and the acoustic power consumed by thermoacoustic conversion. Hence, r_v and r_κ can be considered the acoustic resistances due to the viscous and the thermal-relaxation effects, and g is the acoustic source term which directly shows the effect of thermoacoustic conversion. Their expressions are

$$r_v = \frac{\omega\rho_m \Im - f_v}{A_{gas}\left|1 - f_v\right|^2} \tag{16}$$

$$\frac{1}{r_\kappa} = 1 - \gamma\Im f_\kappa \frac{\omega A_{gas}}{\rho_m a^2} \tag{17}$$

$$g = \frac{f_\kappa - f_v}{1 - f_v(1 - Pr)}\beta\nabla T_m \tag{18}$$

The variations of the three parameters are also shown in Fig. 3. As can be seen, the trend of g, reflected by $\left|g\right| / \nabla T_m$, is very similar to β, because g is proportional to β according to Eq. (18). This is why β is so important to evaluate a working fluid. The curves of r_v and r_κ also show peaks near critical point. All these indicate that the properties of critical point may increase both the thermoacoustic conversion (reflected by $\left|g\right| / \nabla T_m$) and losses (reflected by r_v and r_κ). A similar phenomenon has been observed in simulations for a thermoacoustic engine with the fluid near critical point (Migliorino and Scalo, 2020).

The unique properties of a liquid leads to a huge difference in the acoustic power propagation, with respect to a gas. This is reflected by the large characteristic acoustic impedance $\rho_m a$ of a liquid, which makes the displacement amplitude in a liquid system much smaller than that in a gas system. Specifically, the required p_1 and U_1 to deliver a given amount of acoustic power

satisfies $\dot{E}_2 = \frac{1}{2}\left|p_1\right|\left|U_1\right| \cos\varphi$. In a pure traveling-wave field, $\dot{E}_2 = \frac{A\omega^2}{2}\left|\xi_1\right|^2 \rho_m a$ tells us the required velocity amplitude under a specific characteristic acoustic impedance. In Fig. 4(a), we show the ratio of $\left|\xi_1\right|_{lq} / \left|\xi_1\right|_{he}$, where $\left|\xi_1\right|_{lq}$ and $\left|\xi_1\right|_{he}$ represent the required displacement amplitude to deliver a given amount of acoustic power when liquid propylene and helium are used respectively, with various temperatures and pressures. As can be seen, $\left|\xi_1\right|_{lq}$ is much lower than $\left|\xi_1\right|_{he}$. When $p_m = 10$ MPa and $T_m = 300$ K, the displacement amplitude with propylene is also only 0.2 of that with helium. Consequently, we can envisage a linear motor for driving a liquid like propylene, whose displacement amplitude of the piston is only 0.2 of that for driving helium to deliver a same amount of acoustic power. This is significant to reduce the volume of the linear motor to drive the refrigerator.

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Fig. 3. Properties of propylene under different temperature and pressure. (a) Thermal expansion coefficient. (b) Prandtl number. (c) Density. (d) The ratio of acoustic source term over temperature gradient. (e) Viscous resistance. (f) The reverse of thermal-relaxation resistance.

The heat transfer between the working fluid and the solid wall in heat exchangers is critical in thermoacoustic refrigerators. A poor heat transfer leads to a large ΔT_{fs} (the temperature difference between the working fluid and the solid of heat exchangers), which will significantly dilute the performance of low-temperature-difference thermoacoustic systems (Nsofor et al., 2007, Yang et al., 2022b). The heat transfer in oscillatory flow is a very complicated issue. It is different from the heat transfer in unidirectional flow, because the boundary layer of fluid particles near the wall is much smaller than that in unidirectional flow (Nsofor et al., 2007). There exist some heat transfer correlations fitted from experimental data, but they are only applicable for specific fluids and conditions. Unfortunately, we cannot find any correlation for liquid propylene. As a compromise, a simplified model for estimating the heat transfer in oscillatory flow, which is adopted in the famous thermoacoustic software DELTAEC

(Clark et al., 2007), is used here:

$$\Delta T_{\mathrm{fs}} = \frac{Q}{hA_{\mathrm{gas}}x_{\mathrm{eff}} / r_h} \tag{19}$$

The heat transfer coefficient can be calculated from

$$h = \frac{k}{\min \delta_\kappa, r_h} \tag{20}$$

By assuming $r_h > \delta_\kappa$, which is true in many thermoacoustic systems (Jin et al., 2016, Chi et al., 2021), we reach $h = \sqrt{2\rho c_p \omega k}$. x_{eff} is the effective length of the HX, which can be expressed as

$$x_{eff} = \min l_{hx}, 2\left|\xi_1\right| \tag{21}$$

where $\left|\xi_1\right|$ is the displacement amplitude of the fluid in HX, and l_{hx} is the length of HX. Eq. (21) captures an important distinction between the oscillatory and the unidirectional flow heat transfer: the oscillatory one cannot be enhanced by increasing the length of HX in the streamwise direction if it is larger than $2\left|\xi_1\right|$. To fully facilitate the area of HX, l_{hx} is generally designed to be larger than $2\left|\xi_1\right|$ (Swift, 2002). Therefore, $x_{eff} = 2\left|\xi_1\right|$. Note that Eqs. (19) to (21) oversimplify the heat transfer in the channel. The errors from them should be acceptable in this work because the absolute values of ΔT_{fs} are very low.

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Fig. 4. A comparison between liquid propylene and gaseous helium under different temperature and pressure. (a)Ratio of displacement amplitude. $\left|\xi_1\right|_{lq}$ and $\left|\xi_1\right|_{he}$ represent displacement amplitudes of liquid propylene and gaseous helium respectively. (b)Ratio of heat transfer coefficient. h_{lq} and h_{he} represent heat transfer coefficients of liquid propylene and gaseous helium respectively.

In Fig. 4(b), it is shown that the heat transfer coefficient when liquid propylene is used is around 4 to 6 times higher than that when helium is used; while x_{eff} is about 0.2 of that when helium is used according to Fig. 4(a). Therefore, the temperature difference ΔT_{fs}in the heat exchanger when propylene is used is similar to that when helium is used, with the same r_h, ω, and Q. Even so, the related heat exchangers can be much shorter than that adopting gases, because x_{eff} is only 0.2 of that of helium (see Fig. 4(a)). This can be used to improve the compactness of heat

exchangers, which is one of the most complicated and expensive component in thermoacoustic systems (Swift, 1989b, Migliori and Swift, 1988).

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Fig. 5. Influence of Womersley number on the relative Carnot coefficient of performance. (a) With various $\nabla T_m / \omega$. (b) With various acoustic impedance. $p_m = 10$ MPa.

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Fig. 6. Influence of temperature on the relative Carnot coefficient of performance, with various pressure. $\omega\tau$ is optimized at every point. $|z|$ is fixed at 20 $\rho_m a$. $\nabla T_m / \omega$ is fixed at 9.5 K·s/m.

3.2. Short refrigerator

According to Eq. (11), since the fluid is propylene and the phase difference is $\varphi = 0$, there are five parameters important: $T_m, \nabla T_m / \omega, |z|, \omega\tau$ and p_m. Their influences on the performance of the short refrigerator will be discussed in this section.

In Fig. 5, the values of COP_R with the variation of $\omega\tau$, under different $\nabla T_m / \omega$ or $|z|$ are presented. p_m is fixed at 10 MPa. Similar to the thermoacoustic conversion with gas (Yazaki et al., 1998), Womersley number $\omega\tau$ is a key parameter governing the thermoacoustic conversion with liquid, which is reflected in Eqs. (11), (6b). It represents the ratio between the hydraulic radius over the thermal penetration depth, reflecting a trade-off between the viscous loss and the thermal effect, because it can be rearranged as the ratio of the thermal time scale $\left(r_h^2 / 2\kappa\right)$ to the oscillation time scale $\left(1 / \omega\right)$. If $\omega\tau$ is too large, the thermoacoustic conversion only happens in part of the channel near the wall and the conversion is not efficient because the fluid in the center of the channel is wasted, although the viscous loss may be low; If $\omega\tau$ is too small, the very thin channel will cause severe viscous loss and decrease efficiency (Yazaki et al., 1998). As a result, an optimal $\omega\tau$ exists for highest COP_R in each curve, as shown in Fig. 5(a). Furthermore, the peak COP_R shows a decreasing trend as $\nabla T_m / \omega$ is increased. For lower values of $\nabla T_m / \omega$, the dependence of COP_R on $\omega\tau$ is less sensitive, and the

optimal range of $\omega\tau$ for high COP_R is larger. It helps the whole regenerator to be efficient in a full-scale system, where $\omega\tau$ varies significantly along the regenerator due to the variation of temperature. This is in accordance with the descriptions in Swift (1989b), where it is stated that the liquid fluid only fits for low temperature difference applications.

From Fig. 5(b), it is clear that the acoustic impedance $|z|$ strongly affects COP_R. $|z|$ larger than 20 $\rho_m a$ should be enough for efficient conversion, which is similar to the values suggested in gas traveling-wave thermoacoustic systems (Gardner and Swift, 2003). Otherwise, a sever viscous loss may occur, due to the large velocity amplitude relative to the pressure amplitude. The peak appears near $\omega\tau \sim 0.1$ in all curves, implying that the optimal hydraulic radius of the regenerator weakly depends on $|z|$. Additionally, the values of COP_R can be above 70% if $\omega\tau$, ∇T_m / ω and $|z|$ are properly set, indicating that the thermoacoustic refrigeration with the compressible liquid as the working fluid can be efficient.

The influences of temperature and mean pressure on the performance of the short refrigerator are shown in Fig. 6. Note that $\omega\tau$ is optimized at every point, and $|z|$ is fixed at a relatively large value, so the results here can be considered the theoretical limit of COP_R for a liquid thermoacoustic refrigerator working under similar conditions. The dependence of COP_R on p_m is weak, unless near the critical point. On the other hand, the operating temperature T_m has a significant influence on the performance of the system. COP_R increases with the rise of T_m until reaches a peak. The location of the peak in each curve is near that in Fig. 3, as a trade-off between the gains and the losses. However, at the critical point, COP_R drops significantly towards zero. This tendency indicates that the thermoacoustic refrigeration stops working at the critical point, because of the sudden rise of viscous and thermal-relaxation losses, although $|g|$ is also very high (see Fig. 3). Therefore, a thermoacoustic refrigerator should work away from the critical point, and a mean pressure higher than p_{cr} is a proper choice. Additionally, COP_R of ideal gas (helium) with $p_m = 10$ MPa is also displayed. As can be seen, COP_R of helium gradually decreases due to the increasing viscosity. Compared with liquid propylene, COP_R of helium has a much weaker dependence on T_m, because of the weaker dependence of ideal-gas properties. The comparison shows that

the potential of efficiency with a compressible liquid may be lower than that with an ideal gas.

3.3. Full-scale system

In this section, the performance of the full-scale liquid thermoacoustic refrigerator will be discussed.

In Fig. 7, the values of COP_R^s and Q_c of the system under different $\omega\tau$ and various temperature difference ΔT_m are shown. Obviously, an optimal $\omega\tau$ exists for the highest COP_R^s. The optimal $\omega\tau$ decreases from 0.37 to 0.2 when ΔT_m is increased from 10 K to 40 K. When ΔT_m is increased, COP_R^s shows a weaker dependence on the variation of $\omega\tau$. These are in accordance with the conclusion from the short refrigerator analysis (see Fig. 5). For the air conditioning application where a typical ΔT_m is 25 K, $\omega\tau$ in the range of 0.2 to 0.4 is suggested. The highest COP_R^s is 36% with $\Delta T_m = 25$ K (the corresponding COP^s is 4.1), which is lower than the peak values in the short refrigerator (see Fig. 5). The main reason behind this is that in the full-scale refrigerator, the losses in components including heat exchangers, the resonator and the thermal buffer tube are included, while they are not considered in the short refrigerator. If these losses are ignored, the corresponding COP_R^s can be increased to 42%.

In Fig. 7(b), the corresponding cooling power Q_c is also presented. Q_c rises with the increase of $\omega\tau$, until reaches a peak at around 0.1 to 0.15, then decreases gradually as $\omega\tau$ is further increased. Q_c increases as ΔT_m is decreased, because the acoustic power consumed to pump a specific amount of heat is reduced.

In Fig. 8, the distributions of the acoustic field along the axial direction of the system are presented. Due to the symmetric configuration, only a quarter of the system is demonstrated. This is why the variations of both $ph\left(p_1\right)$ and $ph\left(U_1\right)$ are 90° in Fig. 8(a). For the design of a traveling-wave thermoacoustic system, one of the main aims is to create an acoustic field dominated by traveling-wave components, which is critical to improve thermoacoustic conversion and decrease acoustic power dissipation during the propagation of sound wave (Swift, 2002). This is realized in the presented system. It is shown that the phase difference φ varies from 18° to $-2°$ in the regenerator, while it is in the range of $-11°$ to 20° in other components.

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Fig. 7. Performance of the full-scale system with the variation of Womersley number and different temperature drop . $T_a = 310$ K. $p_m = 10$ MPa. $|p_1|$ is fixed at 0.85 MPa at the ambient heat exchanger.

In Fig. 8(b), both $|p_1|$ and $|U_1|$ decrease significantly in the regenerator, due to the thermoacoustic conversion in it. Obviously, the trends of $|p_1|$ and $|U_1|$ are opposite, because of the existence of standing-wave components. As revealed in Fig. 4, the displacement amplitude $|\xi_1|$, which equals $|U_1| / A_{gas}\omega$, can be very small. In this example, the displacement amplitude in the regenerator is only around 0.3 mm. Further, $|p_1|A_{gas} / |U_1|$ generates the acoustic impedance $|z|$, which is another important index to evaluate the acoustic field inside a traveling-wave thermoacoustic system. Here, $|z|$ is in the range of 15 $\rho_m a$ to 18 $\rho_m a$ in the regenerator, close to the optimal values suggested through the short refrigerator model in Fig. 5.

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Fig. 8. The distributions of acoustic field along the axial direction of the system. (a) Phase angle of p_1, U_1 and their difference. (b) Pressure amplitude and volumetric velocity amplitude. (c) Acoustic power. (d)Temperature. Due to the symmetry of the configuration, only a quarter of the system is presented.

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Fig. 9. (a) Performance of the system with different ambient temperature. (b) Acoustic impedance z_{as} with different ambient temperature. $z_{as} = p_{1,as} / U_{1,as}A_{as}$, where $p_{1,as}$ and $U_{1,as}$ represent the 1st-order pressure and volumetric velocity at the joint of acoustic source and the resonator. A_{as} is the cross-sectional area of the outlet of the acoustic source. $p_m = 10$ MPa. $\omega\tau$ is fixed at 0.27. $|p_1|$ is fixed at 0.85 MPa at AHX. $\Delta T_m = 25$ K.

In Fig. 8(c), the acoustic power $\dot{E}_2$ decreases abruptly in the regenerator, due to the thermoacoustic conversion inside it to pump heat. The only increase of $\dot{E}_2$ appears at the acoustic source, which drives the system. In the acoustic source, the

acoustic power is supplied by adding volumetric velocity according to the Kirchhoff's law, so there also exist sudden changes in $ph(U_1)$, φ and $|U_1|$.

Additionally, the temperature distribution is shown in Fig. 8(d). A temperature drop from 310 K to 285 K happens inside the regenerator due to the thermoacoustic effect. In all heat exchangers and the resonator, temperature is assumed constant.

In summary, the characteristics of the acoustic field in this system are similar to conventional traveling-wave thermoacoustic systems (Jin et al., 2016, Chen and Xu, 2021). Therefore, the adoption of liquid as the working fluid does not alter the basic acoustic features of a conventional thermoacoustic system.

In Fig. 9, the influence of temperature on the performance of the system is presented. From Fig. 9(a), the cooling power Q_c increases dramatically with the rise of T_a, and reaches a peak of 1530 W at $T_a = 375$ K. This can be mainly ascribed to two reasons. The first one is the improved thermoacoustic conversion, which can be reflected by the increased β and $|g|$ with T_m (see Fig. 3). The other non-negligible reason is the increased input acoustic power. Although $|p_1|$ is fixed at 0.85 MPa at AHX, the corresponding acoustic power is increased due to the decrease of characteristic acoustic impedance $\rho_m a$. Specifically, the acoustic power at AHX is increased from 819 W at $T_a = 250$ K to 863 W at $T_a = 319$ K. COP_R^s has a trend similar to Q_c, and it peaks at 357 K with the peak value of 44%. As for COP^s, it can reach 5.9 when T_a is around 360 K. These trends are in accordance with those in the short refrigerator, but the corresponding values of T_a where peaks appear are lower than that in Fig. 6. The reason behind is that the full-scale refrigerator is much more complicated than the short refrigerator, and all components in the system are strongly coupled in terms of acoustic field. Hence, the performance of the system is also affected by the acoustic propagation in components other than the regenerator. Additionally, the hydraulic radius of the regenerator is $10\ \mu m$ to $13\ \mu m$, to satisfy $\omega\tau = 0.27$, and the corresponding thermal penetration depth is $19\ \mu m$ to $24\ \mu m$.

In Fig. 9(b), the acoustic impedance z_{as} is also presented. $z_{as} = p_{1,as} A_{as} / U_{1,as}$, where $p_{1,as}$ and $U_{1,as}$ represent the 1st-order pressure and volumetric velocity at the joint of the acoustic source and the resonator. A_{as} is the cross-sectional area of the

outlet of the acoustic source. As can be seen, both the amplitude $|z_{as}|$ and the phase $ph(z_{as})$ change significantly when $T_a > 330$ K. The sudden changes are caused by the strong dependence of properties of liquid propylene on temperature. Obviously, when the system works in the compressible liquid region, the required impedance z_{as} strongly depends on temperature, indicating that an ordinary linear motor may only fit a narrow range of the temperature variation. An acoustic source with a large adjustable range of acoustic impedance is required to better couple the liquid system.

Another interesting topic is the influence of the trans-critical state on the performance of the system. In Fig. 10, we present the variations of COP_R^s and cooling power Q_c of the system near and in the trans-critical region, with different mean pressures. Since the temperature drop in the system is fixed at 25 K, the temperature range of $364.2\text{K} < T_a < 389.2\text{K}$ covers the critical temperature ($T_{cr} = 364.2$ K). In this range that has been shaded in the figure, the system operates in the trans-critical state. As can be seen, curves of COP_R^s of all mean pressures decrease as T_a approaches towards or enters into the trans-critical region. This trend is increasingly significant when mean pressure is decreased towards the critical pressure. Meanwhile, curves of Q_c show similar trends, although they increase slightly before decrease. The reason behind is that the thermoacoustic refrigeration is deteriorated when the state of the fluid goes towards the critical point, and stops working at the critical point. This has already been revealed in Fig. 6. Therefore, the system should work in the liquid region, rather than the trans-critical region.

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Fig. 10. Performance of the system near and in the transcritical state. (a) The relative Carnot coefficient of performance COP_R^s. (b) Cooling power Q_c. .

4. Conclusions

We have investigated the thermoacoustic refrigeration with compressible liquid as the working fluid based on two models. The first model is the highly-simplified short refrigerator, which can be used to investigate the full potential of the liquid thermoacoustic refrigeration without considering the specific

configuration of the system. The second model is a full-scale acoustic-driven traveling-wave thermoacoustic refrigerator, which gives a more practical estimation on its performance.

In the short refrigerator discussion, the factors affecting the liquid thermoacoustic conversion, including temperature, the ratio of temperature gradient over angular frequency, acoustic impedance and Womersley number, have been examined. By properly setting these parameters, the coefficient of performance can reach as high as 70% of the Carnot limit. This indicates that the thermoacoustic refrigeration with a compressible liquid working fluid has the potential to be as efficient as the conventional gas system. At the critical point, the performance may be ruined by the large losses, so the operation should be far away from the critical point.

In the full-scale system discussion, it can be concluded the acoustical characteristics in a liquid system are similar to those in a gas system, so the basic designs for gas thermoacoustic refrigerators may be also applicable for a liquid thermoacoustic refrigerator. In a typical working condition for the air conditioning application (pumping heat from 285 K to 310 K), the coefficient of performance of the system can reach as high as 4.1, with respect to 36% of the Carnot limit. When pumping heat from 332 K to 357 K, the coefficient of performance can reach 44% of Carnot limit.

Its advantages over the gas system mostly lie on its much smaller displacement amplitude when delivering a given amount of acoustic power, which reduces the displacement amplitude of the motor's piton, and the effective length of the heat exchanger. As a result, both the linear motor and the heat exchanger can be much more compact, so that the mass and the volume of the system can be greatly reduced, as well as the cost. Its potential application is the low-temperature difference refrigeration and heat pumping (air conditioning, electronics cooling, etc.). However, to achieve the very high mean pressure for the proper fluid state, which is higher than that in most classical thermoacoustic systems, a charging system and a more stringent design on the pressure resistance are needed. This disadvantage is acceptable in a lab environment, but might be a hindrance for the further development of such systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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